Electric current

Introduction

Electric current is the rate of flow of charge carriers. When charge carriers (electrons, ions) move through a material, they create a current. Direction of current:

- Conventional: from + to − (use this in exams)
- Electron flow: from − to +

Charge

Charge is what is actually flowing

$$\boxed{Q = Nq}$$

Where:

Q − total charge, unit: coulombs (C)
N − number of charge carriers
q − charge per carrier. For electrons: $q = e = 1.60 \times 10^{-19}$C

$N = n \times V$ (Volume; number density of charge carriers (n))

Calculating current

Electric current is the rate of flow of charge carriers, calculated using:

$$\boxed{I = \frac{Q}{t}}$$

Where:

I − current (A), unit: amperes (A)
Q − charge (C)
t − time (s)

You could also calculate charge using: $\boxed{Q = It}$
Using this equation we can define coulomb as ampere second

Current in current-carrying conductor ($I = nqAv$)

Derivation:

$$I = \frac{Q}{t} = \frac{Nq}{t} = \frac{nVq}{t} = \frac{nq \times L \times A}{t} = nqAv$$

$$\boxed{I = nqAv}$$

Where:

I − Current (A)
n − Number density of charge carriers (m^{-3})
A − Cross-sectional area (m^2)
v − Drift velocity (ms^{-1}), avg velocity of charge carriers
q − Charge per carrier (C)

Consider a wire with cross-sectional area A where charge carries flow at an average velocity of v ($v = \frac{L}{t}$)

Energy Transfers in Circuits (p.d. and e.m.f.)

Potential difference (p.d.) is energy transferred from electrical to other forms (heat, light, sound, etc.) per unit charge. It is measured across two points (component) − it's the "drop" in energy as charge goes through.

$$\boxed{V = \frac{W}{Q}}$$

Where:

V − potential difference, unit: volts (V)
W − energy transferred (J)
Q − charge (C)

Electromotive force (e.m.f.) is the energy transferred from other forms/chemical to electrical per unit charge as charge passes through a source (battery, generator)

$$\boxed{E = \frac{W}{Q}}$$

Where:

E − e.m.f., unit: volts (V)
W − energy transferred (J)
Q − charge (C)

When asked to distinguish between e.m.f. and p.d.:
Compare the energy conversion:
- e.m.f. → other forms → electrical
- p.d. → electrical → other forms

Resistance

Resistance is a measure of how difficult it is for current to flow through a component, defined as potential difference / current.

$$\boxed{R = \frac{V}{I}}$$

$\boxed{V = IR \quad \text{or} \quad I = \frac{V}{R}}$

$V \propto R$ (if I constant)
$V \propto I$ (if R constant)
$I \propto \frac{1}{R}$ (if V constant)

Where:

R − resistance, unit: ohms (Ω)
V − potential difference across component (V)
I − current through component (A)

Resistance opposes the flow of charge. It is caused by collisions between charge carriers (electrons) and the atoms of the conducting material. These collisions transfer energy from the carriers to the atoms (usually as heat).

Resistivity

The resistance of a wire depends on three factors:

Factor	Effect
Length (L)	Longer wire → higher resistance
Cross-sectional area (A)	Thicker wire (larger A) → lower resistance
Material	Different materials oppose current by different amounts (resistivity)
Temperature	May affect resistance (explained in I-V characteristics)

Resistivity Equation

$$\boxed{R = \frac{\rho L}{A}}$$

Other forms:

- $\rho = \frac{AR}{L}$
- $L = \frac{AR}{\rho}$
- $A = \frac{\rho L}{R}$

Where:

R − resistance (Ω)
ρ − resistivity (Greek letter rho), unit: ohm-metre (Ω m)
L − length of wire (m)
A − cross-sectional area (m^2)

Resistivity (ρ) is a property of a material. It tells you how strongly that material opposes current, regardless of its shape or size. (resistance for a wire of length 1 m per area $1m^2$)

I−V Characteristics

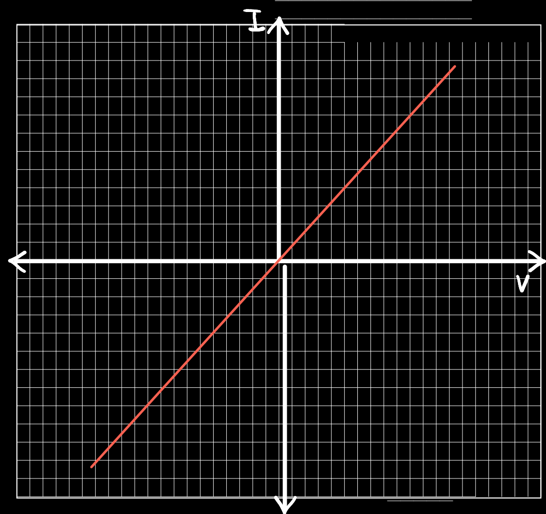
Ohm's law: For a metallic conductor at constant temperature, current is directly proportional to potential difference ($V \propto I$).

Ohmic vs Non-Ohmic Conductors

−Ohmic: its resistance remains constant regardless of current and p.d → Obeys $V \propto I$
−Non-Ohmic: Its resistance changes as current changes → Does not obey $V \propto I$

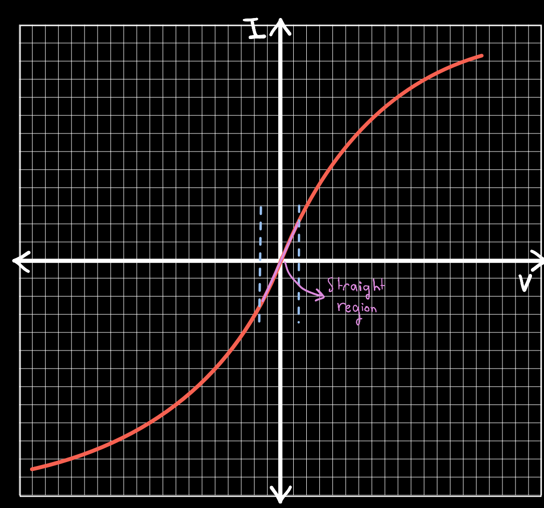
Resistor

OR (Metallic Conductor at constant temperature)



Explanation:
- Straight line through origin
- Current is directly proportional to potential difference
- Resistance is constant (obeys Ohm's law)
- Temperature must remain constant (only in case of Metallic Conductor while for resistor it is constant at all tempretures)

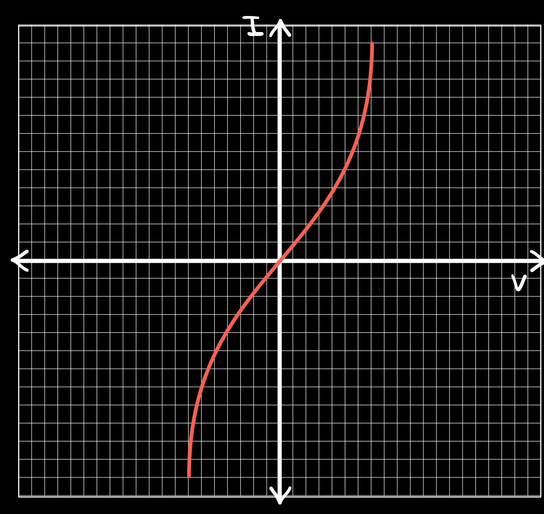
Filament Lamp



Explanation:
As voltage increases → current increases → temperature rises →
Ratio V / I increases → Resistance increases
Does not obey Ohm's law overall (except near origin)

NOTE: Near the origin, gradient is constant → behaves ohmic (temperature not yet increased)

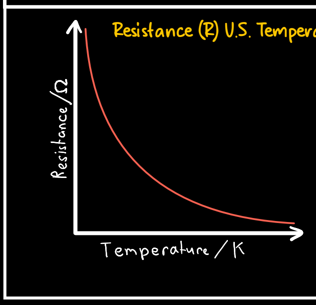
Thermistor



Explanation:
As voltage increases → current increases → temperature rises → Ratio V/I decreases → Resistance decreases
Does not obey Ohm's law overall

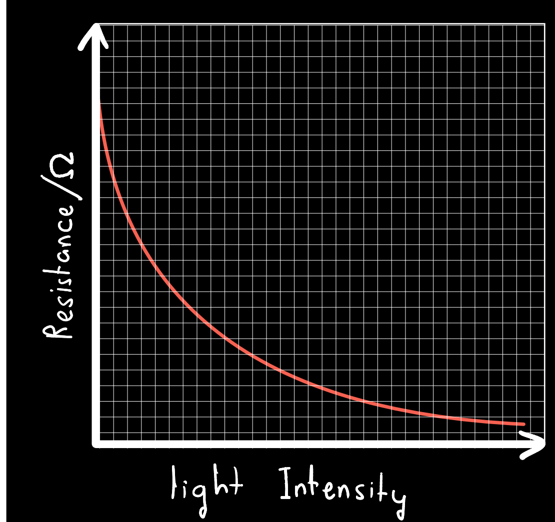
Characteristic:
Resistance depends on tempreture
- High tempreture → low resistance
- Low tempreture → high resistance

Resistance (R) V.S. Temperature (kelvin)



Explanation:
Curve decreases exponentially
As temperature increases → resistance decreases
Does not touch the x-axis or y-axis (when T is in kelvin)
Exception: If temperature is in Celsius, the graph touches the y-axis at 0°C (but not x-axis)

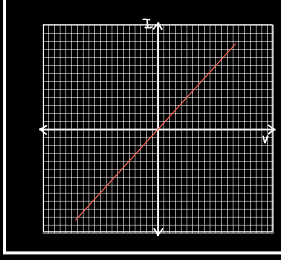
LDR (Light-Dependent Resistor)



Explanation:
Curve decreases
As light intensity increases → resistance decreases
Touches Y-axis BUT Does not touch X-axis (resistance never reaches zero)

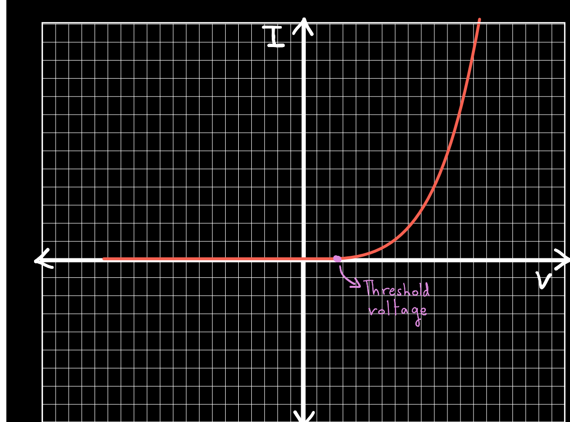
Characteristic:
Resistance depends on light intensity
- High light intensity → low resistance
- Low light intensity (dark) → high resistance

I−V at Constant Light Intensity



Explanation:
At constant light intensity, resistance is fixed
Current ∝ voltage (ohmic)
Obeys Ohm's law

Diode



Explanation:
- resistance = V / I
- Very high/infinite resistance for negative voltages up to threshold voltage
- Resistance decreases as voltage increases (past threshold voltage)

Exam tip: Mention the exact voltage threshold voltage value (e.g. 0.6 V)

Threshold voltage:
Below this voltage: diode has very high resistance, almost no current flows
At/above this voltage: resistance drops sharply, current increases rapidly

Power

Power is the rate of energy transfer.

$$\boxed{P = \frac{W}{t}}$$

Where:

P − power, unit: watts (W)
W − energy (J)
t − time (s)

Electrical Power

$V = \frac{W}{Q} \rightarrow W = VQ$

$I = \frac{Q}{t} \rightarrow t = \frac{Q}{I}$

$P = \frac{W}{t} = VQ \div \frac{Q}{I} \Rightarrow \boxed{P = VI}$

Other forms:

$\boxed{P = I^2R}$ ($IR = V$)

$\boxed{P = \frac{V^2}{R}}$ ($\frac{V}{R} = I$)

Appliance Power and Voltage Rating

Every electrical appliance has a rating (Power, Voltage)
- Voltage rating: The p.d. the appliance is designed to operate at.
- Power rating: The power the appliance consumes when operated at its rated voltage.

Example: A lamp marked "60 W, 240 V" will:
Consume 60 W of power when connected to 240 V
Draw a current I = P/V = 60/240 = 0.25 A
Have resistance R = V/I = 240/0.25 = 960 Ω

If connected to a different voltage:
Lower voltage → lower power (bulb dimmer)
Higher voltage → higher power (bulb brighter, may burn out)

Exam tip:
If asked to define a unit, give its definition in terms of other units:
E.g. Coulomb (C) ⟹ Ampere second (A·s)
Volt (V) ⟹ Joule per coulomb (J/C)
Ohm (Ω) ⟹ Volt per ampere (V/A)
Watt (W) ⟹ Joule per second (J/s) or volt × ampere (V·A)
Key pattern: Unit definition = its formula rearranged with units.